FARID KARIMLI

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EDUCATION

Boston University, Faculty of Computing and Data Sciences Boston, MA
Ph.D., Data Science Started Sept 2025

M.S., Data Science, GPA 3.8 Jan 2025
Thesis: *Advancing Human-AI Collaboration: Multimodal Systems for Accessibility, Research, and Learning*

B.A, Computer Science with Minor in Economics, GPA 3.7 May 2023

PUBLICATIONS & WORKING PAPERS

- Farid Karimli & Onur Okuducu (in preparation). "*Improving Fine-grained Video Understanding with Pose-based Self-Supervised Learning.*"
- Farid Karimli and Margrit Betke. 2025. *Enhancing Text Entry for Users with Motor Impairments: Design and Evaluation of KeyGlide.* In Proceedings of the 18th ACM International Conference on PErvasive Technologies Related to Assistive Environments (PETRA '25). Association for Computing Machinery, New York, NY, USA, 343–348. https://doi.org/10.1145/3733155.3733218.
- Farid Karimli, Hao Yu, Emmanuel Akosah, Srishti Jain, Margrit Betke, and Wenxin Feng. 2024. *Demonstration of CameraMouseAI: A Head-Based Mouse-Control System for People with Severe Motor Disabilities*. The 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), October 27-30, 2024, St. John's, NL, Canada. 6 pages. https://doi.org/10.1145/3663548.3688499

ONGOING RESEARCH

- **Subject-driven Grounded Image Generation** – Developing a method for controllable interior scene generation by grounding diffusion models with concrete objects from a catalogue; exploring evaluation metrics for scene realism and faithfulness to constraints.
- **Fine-grained Video Understanding via Self-Supervised Learning** – Leveraging multimodal signals (pose trajectories and temporal event cues) to enhance fine-grained activity recognition; investigating hybrid SSL + I2V approaches to improve representation learning for complex human motion.

RESEARCH & WORK EXPERIENCE

Vincer Cambridge, MA
Full-Stack AI Engineer Feb 2025 – July 2025

- Developed web clients using React, Vite, and Supabase, enabling seamless user interaction with the AI-powered sales productivity platform.
- Designed data pipelines to connect the frontend with a FastAPI backend deployed on AWS with Docker.
- Integrated real-time video meeting functionality using Dyte, allowing mentors and sales professionals to connect directly within the product.
- Built agentic AI workflows with LangGraph to power the "AI Concierge," orchestrating multi-agent interactions that guided user conversations, routed tasks, and flagged high-risk cases beyond AI scope.

Gardos Lab and Image & Vision Computing Group Boston, MA
Computer Vision and NLP Research – Advised by Dr. Margrit Betke and Dr. Thomas Gardos Jun 2023 – Dec 2024

- Enhanced facial gesture recognition throughput by 70% using SSD models and mediapipe on CameraMouse, enabling faster clicking interaction.
- Conducted iterative software testing with users with motor impairments to refine interface usability.
- Improved complex PDF visual data parsing and understanding using LLM-powered data pipelines.
- Boosted CLIP model inference throughput on 200K+ images by 200% using PyTorch DDP
- Integrated CLIP with a SWIN backbone, improving zero-shot prediction by 30% on 100+ herbarium species.

BU CS Department Boston, MA
Teaching Assistant Sep 2022 – Dec 2024

- Most recently taught graduate data science, covering topics in clustering, classification and regression.
- Previously taught introductory CS in weekly labs, held 4 hours of office hours and graded 400+ exams.
- Mentored students on programming fundamentals, personal projects and problem sets.

Software Application and Innovation Lab Boston, MA
Software Engineering Intern May 2022 - Aug 2022

- Created an interactive network graph using D3.js, enabling linguists to visualize ASL lexicon relationships.
- Streamlined backend functionality using API payload validation tests and resolving dependency issues in Android applications.
- Deployed a real-time operations dashboard using ClickUp API, providing actionable insights for hiring workflows and reducing data query time by 30%.

PROJECTS

Edubotics Jun 2024

- Open-source library for developing academic LLM-powered chatbots for course help.
- Created a modular RAG system for content retrieval on lectures and assignments, increasing accuracy by 63%.
- Built a GPT4o-enabled data extraction pipeline, enhancing PDF parsing accuracy and response relevance.
- Applied OOP design patterns in Python to maintain modularity, simplifying extensibility of components.

AirMouse | DEMO

- Developed a desktop vision-based application enabling users to control the mouse pointer using hand gestures.
- Leveraged Mediapipe to detect and hand landmarks for precise cursor movement and gesture recognition.
- Implemented custom gesture detection algorithms for left-click, right-click, and drag-and-drop functionalities, achieving 95% accuracy in real-time testing.

CameraMouseAI

- Implemented an interface that enables folks to control computers using head movements and facial gestures.
- Added facial gesture recognition for mouse command using real-time 3D facial landmark detection in Mediapipe.
- Collaborated with researchers to integrate optical flow-based tracking, enabling precise control.

SKILLS

Programming Languages: Python, Java, Javascript, TypeScript, R.
Tools/Frameworks: Numpy**,** Pandas, PyTorch**,** Google Cloud, AWS, MySQL, Docker, REST APIs, React
Knowledge Areas: Machine Learning, Deep Learning, Computer Vision, LLMs, Big Data Processing, Web Applications